National University of Computer and Emerging Sciences
FAST School of Computing
Fall 2022
CS3002 Information Security
(Section # BCS-7H)

QUIZ # 1 GRADED **Date: September 15, 2022**

Time Allowed: 30 minutes **Maximum Marks: 12**

Instructions: Attempt all questions in sequence. Please write your answers briefly and precisely. Unnecessary or irrelevant details will not add anything to your credit.

1. Consider an automated cash deposit machine in which users provide a card or an account number to deposit cash. Give examples of confidentiality, integrity, and availability requirements associated with the system. (1 + 1 + 1 = 3 marks)

Answer:

Confidentiality:

By confidentiality means only authorized persons have access to the account information. By providing your account number, and your details through card, and finally entering your correct PIN. It's the PIN number giving the sole authorization over your account. Now you have the access to your account and the choice to withdraw or deposit funds or check your balance. Failing in authorization would cause someone else to have access to your account and the balance will be wiped off sooner or later.

Data Integrity:

Data Integrity in the sense, No one in the middle(a malicious person) has access to information between the Cash Deposit Machine (CDM) to the host server. It should maintain its unaltered original form. In a real sense – Details you've given in CDM such as your account number, your PIN, personal information is able to reach its host server without alteration in data or eavesdropping by someone. This ability of untouched or unaltered form communication between CDM and host server is what we call Data integrity.

If somehow failed in data integrity, which would have the highest level of consequences for the error. If some third-party bad actor listens to your account credentials, now he has every information to reach your account and clean your bank balance-so data integrity should be given the highest degree of importance.

Availability:

When you enter all your credentials and server would give you the required information or service such as bank balance or deposit/withdrawal of funds — The property of an CDM (in this case) being able to provide all the requested services without delay is what we call availability. Being unable to provide service for an authorized customer would give a bad experience for the customer, so the consequences are more on the service provider than the customer.

Diversity in answers is possible (logical answers needed)

2. The input given to final permutation in Data Encryption Standard (DES) is 0x 0002 0000 0000 0001. Find the output. Explain in two sentences (strictly) how did you get to the answer? (1 + 1 = 2 marks)

Final Permutation							
40	08	48	16	56	24	64	32
39	07	47	15	55	23	63	31
38	06	46	14	54	22	62	30
37	05	45	13	53	21	61	29
36	04	44	12	52	20	60	28
35	03	43	11	51	19	59	27
34	02	42	10	50	18	58	26
33	01	41	09	49	17	57	25

Solution:

Binary Equivalent is 001000000000000000000000000000000000000001. Hence, the resultant hexadecimal will be:

0x 02 10 00 00 00 00 00 00

3. Show the result of passing 110010 and 010111 through S-box 1 in the Data Encryption Algorithm (DES). (1 + 1 = 2 marks)

S-box 1

	0	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*
0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07
1	00	15	07	04	14	02	13	10	03	06	12	11	09	05	03	08
2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00
3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13

Figure 1: S-Box 1 of DES Algorithm

Solution:

110010 -> 10 is 2rd row and 1001 is 9th column hence we get 12 of decimal that is 1100 in binary.

010111 -> 01 is 1st row and 1011 is 11th column hence we get 11 of decimal that is 1011 in binary.

4. What kind of a benefit or a security requirement does a Permutation-Box (P-Box) provide in DES? Give reason in one or two sentence(s) (strictly). (1 + 1 = 2 marks)

Solution:

Diffusion of bits. By changing the location of in every round, the P-Box ensures the effect of one cipher text bit computed from several plaintext bits and vice versa.

5. What does a cryptanalytic attach exploit and what does it try or attempt to deduce? How it is different from brute-force attack? (1 + 2 = 3 marks)

Answer:

It exploits the characteristics of the algorithm to attempt to deduce a specific plaintext or the key being used. It is different because brute-force uses every possible key however cryptanalytic attack relies on known characteristics.